Bell Test Project: Experimental Verification of Quantum Nonlocality

Jiaxin Pan (Group Leader & Introduction)
Yubo Wan (Theory)
Xincheng Zhang (Experiment)
Fan Ye (Data Analysis)
Chenya Zhan (Applications)

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1 Introduction

1.1 Historical Context: The Einstein-Podolsky-Rosen Paradox and Bell's 1964 Paper

The Einstein-Podolsky-Rosen (EPR) paradox, first introduced in their groundbreaking 1935 publication "Can Quantum-Mechanical Description of Physical Reality be Considered Complete?" by Albert Einstein, Boris Podolsky, and Nathan Rosen [1], remains a cornerstone in the ongoing discourse surrounding the interpretation of quantum mechanics. This thought experiment emerged from a deep-seated concern among its authors regarding the completeness of the quantum mechanical framework in its description of physical reality. EPR posited that quantum mechanics failed to account for certain "elements of reality"—physical quantities that they believed possessed definite values independent of measurement—and proposed the possibility of a more comprehensive theory that would incorporate these missing elements, potentially through the introduction of "hidden variables."

The crux of the EPR paradox lies in their "criterion of reality": "If, without in any way disturbing a system, we can predict with certainty (i.e., with probability equal to unity) the value of a physical quantity, then there exists an element of reality corresponding to that quantity." To illustrate their argument, EPR considered a system of two entangled particles. For instance, imagine two particles whose positions and momenta are correlated due to an initial interaction, and subsequently separated by a significant distance. By precisely measuring the position of the first particle, one can predict with absolute certainty the position of the second particle without any direct measurement or physical disturbance. According to the EPR criterion, this predictability implies that the position of the second particle must be an element of reality. Similarly, by measuring the momentum of the first particle, the momentum of the second can be predicted with certainty, leading to the conclusion that its momentum is also an element of reality. However, a central tenet of quantum mechanics at the time was that position and momentum are incompatible observables, meaning they cannot simultaneously possess definite values for a single quantum system. This apparent contradiction led EPR to conclude that quantum mechanics, in its current formulation, provides an incomplete description of physical reality, as it fails to account for these seemingly definite elements.

The EPR argument was deeply rooted in the intuitive principles of locality and realism. Locality, in this context, posits that no influence or information can travel faster than the speed of light, thus precluding any instantaneous effect on one particle due to a measurement performed on the other if they are spatially separated. Realism, as envisioned by EPR, suggests that physical properties have definite, pre-existing values, independent of the act of measurement. Einstein, in particular, expressed his profound discomfort with the implications of quantum mechanics that suggested "spooky action at a distance," where entangled particles appeared to influence each other instantaneously across vast distances, potentially violating the fundamental tenets of his theory of relativity. He harbored the hope that a more complete, deterministic, and local theory could be formulated to provide a more satisfying explanation of the observed phenomena.

In 1964, John Stewart Bell responded to the challenge posed by the EPR paradox with his seminal paper "On the Einstein Podolsky Rosen Paradox" [2]. Bell's work was directly inspired by the EPR argument and aimed to investigate the possibility of constructing a local hidden variable theory that could fully reproduce the predictions of

quantum mechanics. His remarkable achievement was the derivation of a mathematical inequality, now widely known as Bell's inequality. This inequality sets a quantifiable limit on the strength of correlations that can exist between measurements performed on two spatially separated particles if their outcomes are governed by local hidden variables. Crucially, Bell demonstrated that quantum mechanics predicts correlations for certain entangled states that can violate this inequality. This profound result implied that quantum mechanics is fundamentally incompatible with the joint assumptions of local realism.

1.2 Quantum Entanglement: The Heart of Quantum Nonlocality

Quantum entanglement stands as one of the most counterintuitive yet fundamental phenomena in the realm of quantum mechanics. It describes a unique type of correlation that can exist between two or more quantum particles, linking their fates together in a manner that transcends the classical understanding of independent objects. When particles become entangled, their quantum states become intertwined, and they can no longer be described as having separate, independent states; instead, they must be considered as a single, unified quantum system [3].

The properties of entangled particles exhibit correlations that are far stronger and more peculiar than any correlations observed in the classical world. For example, a system with zero total spin that decays into two spin-1/2 particles will exhibit perfect anti-correlation of their spin measurements along the same axis. This non-classical correlation defies the expectations of local hidden variable theories and forms the basis of Bell's inequality tests.

1.3 Bell's Theorem: Challenging Classical Locality and Realism

Bell's Theorem provides a rigorous mathematical framework to examine the fundamental conflict between the predictions of quantum mechanics and the intuitive classical worldview known as local realism. The theorem demonstrates that if quantum predictions about entangled particles are correct, then at least one of the assumptions of locality or realism must be abandoned. The key result is the derivation of Bell inequalities, which place upper bounds on the correlations possible under local hidden variable models.

The Clauser-Horne-Shimony-Holt (CHSH) inequality [4] is a practical and widely used form of Bell's inequality, formulated for feasible experimental implementation. Quantum mechanics predicts violations of this inequality for appropriately chosen measurement settings, providing a testable means to refute local realism.

Characteristic	Brief Description	Relevance to Bell Test
Superposition	Quantum systems can exist in multiple states simultaneously.	Entangled states are often superpositions of multi-particle states, leading to correlations upon measurement.
Correlation	Measurement outcomes of entangled particles are strongly linked, even over large distances.	Bell's inequality tests the limits of correlations achievable under local realism against the stronger correlations predicted by quantum mechanics for entangled particles.
Nonlocality	Correlations appear instantaneous, suggesting influences faster than light (though not usable for faster-than-light communication).	Bell's Theorem and its experimental verification provide evidence for this non-local connection, challenging the principle of locality.
Non-separability	Entangled particles cannot be described by independent quantum states; they form a single quantum system.	This interconnectedness is fundamental to the strong correlations observed and the violation of Bell's inequality.

Table 1: Key Characteristics of Quantum Entanglement and Their Relevance to the Bell Test

1.4 Roadmap of the Report

This project aims to experimentally demonstrate quantum nonlocality through Bell test measurements and theoretical analysis. The subsequent sections of this report are structured as follows:

- **Section 2: Theoretical Background** — Explains the conceptual framework, including local realism, quantum nonlocality, Bell's theorem, and potential loopholes.
- **Section 3: Experimental Setup** — Describes the details of our experimental apparatus, measurement configurations, and the implementation of the Bell test.
- **Section 4: Data Analysis** — Presents the analysis of experimental data, the calculation of correlation functions, derivation of the CHSH inequality, and the interpretation of results.
- **Section 5: Applications** — Discusses modern applications of Bell inequality violations, such as quantum teleportation, quantum computing, and quantum metrology.

This structure reflects the logical progression from theoretical motivations to experimental validation and practical significance of the Bell test.

2 Theoretical Background

2.1 Local Realism

2.1.1 Summary

In the early 20th century, physicists generally accepted **local realism** as the foundation of classical physics.

2.1.2 Locality

The cause of a physical change must be local. According to the theory of relativity, no influence can travel faster than the speed of light. Thus, "action at a distance" with superluminal speed does not exist.

2.1.3 Realism

The properties of physical objects are determined prior to measurement and remain independent of observation.

2.2 Quantum Nonlocality

2.2.1 Summary

The Copenhagen interpretation, led by Niels Bohr, challenged local realism by proposing **quantum nonlocality**—a phenomenon where entangled particles exhibit instantaneous correlations, regardless of spatial separation.

2.2.2 Nonlocality

Instantaneous correlations exist between entangled particles, independent of the distance between them, apparently faster than the speed of light.

2.2.3 Unrealism

The properties of quantum particles are not determined until the moment of measurement. Prior to measurement, particles exist in a superposition of all possible outcomes.

2.3 Heisenberg Uncertainty Principle

The Heisenberg uncertainty principle expresses a fundamental limit on the precision with which certain pairs of physical properties can be simultaneously known:

$$\Delta x \, \Delta p \geq \frac{\hbar}{2}, \tag{1}$$

where Δx and Δp are the standard deviations of position and momentum, respectively, and $\hbar = \frac{h}{2\pi}$ is the reduced Planck constant.

This principle implies that precise knowledge of a particle's position necessarily implies uncertainty in its momentum, and vice versa.

2.4 Einstein–Podolsky–Rosen (EPR) Paradox

Consider a pair of entangled particles. Measuring the position or momentum of the first particle allows precise prediction of the corresponding property of the second particle, even at a distance. EPR argued that this suggested the incompleteness of quantum mechanics, since no action should instantaneously affect a distant system according to relativity.

2.5 Local Hidden Variable Theories (LHVT)

To reconcile locality and realism, Einstein and colleagues proposed the existence of **hidden variables** that determine measurement outcomes but remain unobserved. Under LHVT, if these hidden variables were fully known, quantum systems would be perfectly predictable and superposition would be unnecessary.

2.6 Milestones in the History of Bell's Inequality

2.6.1 The Original Bell's Theorem

In 1964, J. S. Bell published his landmark paper [2], introducing the Bell inequality to test the validity of LHVT against quantum mechanics. Assuming LHVT with hidden variables λ, the correlation function between measurements along directions a and b satisfies:

$$1 + P(b, c) \geq |P(a, b) - P(a, c)|. \tag{2}$$

Here, $P(a, b)$ is the expectation value of the product of measurement outcomes along directions a and b, and $A(a, \lambda), B(b, \lambda) = \pm 1$ are the predetermined measurement results depending on λ.

The derivation begins from:

$$P(a, b) = \int A(a, \lambda) B(b, \lambda) \rho(\lambda)\, d\lambda, \tag{3}$$

where $\rho(\lambda)$ is the probability distribution of hidden variables.

By manipulating these terms and applying the absolute value inequality, one arrives at Bell's inequality. Experimental violation of this inequality implies the failure of local hidden variable models.

2.6.2 The CHSH (Clauser-Horne-Shimony-Holt) Inequality

To make Bell's test experimentally feasible, Clauser, Horne, Shimony, and Holt reformulated the inequality as:

$$S = |E(a, b) - E(a, b') + E(a', b) + E(a', b')| \leq 2, \tag{4}$$

where $E(a, b)$ denotes the expectation value of joint measurements along directions a and b. Violation of this inequality (i.e., $S > 2$) indicates quantum nonlocality.

2.7 Potential Loopholes in Bell Tests

- **Locality loophole:** Measurement settings at one site might influence the outcome at the other site.
- **Detection loophole:** Not all particles are detected, potentially biasing the observed correlations.
- **Coincidence loophole:** Inappropriate setting of coincidence timing windows may lead to biased results.
- **Memory loophole:** Measurement devices may remember previous settings and outcomes.
- **Many-worlds loophole:** Interpretations like the many-worlds view suggest deterministic parallel outcomes, though this remains unprovable.
- **Superdeterminism:** A philosophical view that all events are predetermined, including choices of measurement settings. This hypothesis is also unprovable.

2.8 Historical Experiments and the Closing of Loopholes

The earliest experimental attempts, such as those by Clauser and Freedman, and later by Aspect in 1981, aimed to test Bell's inequality but were limited by locality and detection loopholes.

In 2015, experiments by Shalm *et al.* [5] and Hensen *et al.* [6] achieved loophole-free Bell tests by improving detector efficiency and spatial separation, marking a decisive moment in the history of quantum mechanics.

2.9 GHZ (Greenberger–Horne–Zeilinger) States

GHZ states extend entanglement tests to systems of three or more particles. A three-particle GHZ state is:

$$|GHZ\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle). \tag{5}$$

Unlike Bell states, GHZ states exhibit even stronger contradictions with local realism, providing robust evidence against LHVT.

2.10 Modern Applications of Bell's Inequality

- **Quantum random number generation (QRNG):** Certified true randomness without hidden variables.
- **Quantum communication and networking:** Bell inequality violations certify entanglement in quantum networks.
- **Device-independent quantum key distribution (DI-QKD):** Security based on entanglement verification, even if devices are untrusted.

3 Experimental Setup

3.1 Optical Device

1. Laser Driver

Role: Provides stable power to the laser.
Principle: Converts electrical energy into optical energy to drive the laser's operation. Ensures the laser emits light at a stable power and wavelength.

2. Laser

Role: Generates pump light to excite the BBO crystal and produce entangled photon pairs.
Principle: Utilizes stimulated emission to produce coherent light. The laser emits light at a specific wavelength, which is used to pump the BBO crystal.

3. Filter

Role: Filters out unwanted wavelengths to ensure the purity of the pump light.
Principle: Allows only specific wavelengths to pass through while blocking others. This ensures that the pump light has a high degree of monochromaticity.

4. Polarizer

Role: Filters out specific polarization directions, allowing only light in a particular polarization direction to pass.
Principle: Utilizes the polarization properties of light. Light with a specific polarization direction can pass through the polarizer, while light in other directions is absorbed or reflected.

5. Half-Wave Plate (HWP)

Role: Changes the polarization direction of light.
Principle: Introduces a phase difference of π (half a wavelength) between the two orthogonal polarization components of light. This causes the polarization direction of the light to rotate. The rotation angle is twice the rotation angle of the HWP's fast axis.

6. Quarter-Wave Plate (QWP)

Role: Introduces a phase difference of $\pi/2$ (a quarter of a wavelength) between the two orthogonal polarization components of light.
Principle: Converts linearly polarized light into circularly or elliptically polarized light, or vice versa. This is useful for preparing specific types of entangled states.

7. BBO Crystal

Role: Generates entangled photon pairs through spontaneous parametric down-conversion.
Principle: When pumped with light, the BBO crystal converts a high-energy photon into two lower-energy photons (signal and idler) that are entangled in polarization.

8. Single-Photon Detector

Role: Detects individual photons and records their arrival time and position.
Principle: Utilizes the photoelectric effect. When a photon hits the detector, it generates an electrical signal that can be recorded and analyzed.

9. Coincidence Counter

Role: Records events where two detectors detect photons simultaneously.
Principle: Compares the detection signals from two single-photon detectors. If the signals arrive within a specific time window, it records a coincidence event. This helps in identifying correlated photon pairs.

10. Collimator Adjuster

Role: Adjusts the position and direction of optical elements in the optical path.
Principle: Ensures the accurate alignment of optical elements. Proper alignment is crucial for the efficient generation and detection of entangled photons.

3.2 Experiment Steps

3.2.1 Establishment and adjustment of the optical path

- Click "Experiment Projects" in the right menu of the software interface and select "Bell Inequality Test".
- Turn on the power supply button of the Laser Driver.
- Turn on the power supply button of the coincidence counter and the single-photon detector. Tune the coincidence time window to be 1.5 ns and the delay to be 9 ns by the following operation. After clicking the arrows in the center of the round bottom in the left corner, the cursor will move to the first number in the window menu. We can tune the window time by tuning the arrow of up and down. The cursor will move to the decimal if the right arrow is clicked in the round button.
- Rotate the faster axis of HWP-405nm to 22.5 degree, and keep the faster axis of HWP-810 and QWP-810 nm in two measurement paths as zero, then check the OUT of the Coincidence Count to make sure the rate is 5000-8000. If the count rate is less than 5000, slightly adjust the button on two collimators to make the OUT to be the maximum (about 5000-8000).
- Adjust the faster axes of two HWP-810nm in the measurement path to 22.5 and -22.5 degree, respectively. Tune the horizontal rotation angle of the QWP(z)-405nm by tuning the bottom on the left rather than the middle one to make the OUT of the coincidence count to be the minimum. The corresponding quantum state will be

$$|\varphi\rangle = \frac{1}{\sqrt{2}} \left(|HH\rangle - |VV\rangle \right) \tag{6}$$

3.2.2 Test of the entanglement feature: Bell Test

In this experiment, we utilize the setup of measurement basis composed of wave plate and polarizers. Keep the polarizer to be stationary and rotate the half wave plates. Therefore, we mainly manipulate the half wave plates in the two measurement paths during the experiment.

- Rotate the corresponding half wave plates following the angles in the table and record the corresponding coincidence counts (Please be noted to use the angle in the brackets highlighted in red).
- Open "(Bell's Inequality Test)" in the "(Data Processing)"
- Calculate S value. If the calculated S is larger than 2, it is indicated that the light source is entanglement.

HWP-a / HWP-B	-22.5 (-11.25)	67.5 (33.75)	22.5 (11.25)	112.5 (56.25)
-45(-22.5)	6229	1076	1059	6384
45(22.5)	1086	6203	6222	1069
0(0)	6418	1054	6187	1076
90(45)	1029	6087	1081	1076
S Value				2.8308

Figure 1: Coincidence counts for various HWP-a and HWP-B settings. The angles in red indicate actual rotation angles of the HWPs.

Note: Angles in the table is the polarization of photons, which is double of the rotation angle of HWPs. Therefore, the actual rotation angle of HWPs should be number in the brackets highlighted in red.

3.2.3 Principle of generating entangled photon pairs

The principle of generating entangled photon pairs using a pump light source and a BBO (-barium-borate) crystal is based on the process of Spontaneous Parametric Down-Conversion (SPDC).

Spontaneous Parametric Down-Conversion (SPDC)

In SPDC, a high-energy pump photon is converted into a pair of lower-energy photons (signal and idler) within a nonlinear crystal like BBO. This process must satisfy both energy and momentum conservation laws:

$$\omega_p = \omega_s + \omega_i \quad \textit{(Energy Conservation)} \tag{7}$$

$$\mathbf{k}_p = \mathbf{k}_s + \mathbf{k}_i \quad \textit{(Momentum Conservation)} \tag{8}$$

where ω_p, ω_s and ω_i are the angular frequencies of the pump, signal, and idler photons, respectively, and $\mathbf{k}_p$, $\mathbf{k}_s$, and $\mathbf{k}_i$ are their corresponding wave vectors.

BBO Crystal Characteristics

BBO crystals are uniaxial and exhibit birefringence, meaning they have two principal refractive indices: ordinary (n_o) and extraordinary (n_e). This property is crucial for phase matching, which ensures that the generated photon pairs are coherent. The refractive index for light polarized perpendicular to the crystal's optic axis is n_o, while for light polarized parallel to the optic axis, it is n_e.

Phase Matching

Phase matching is essential for efficient SPDC. In BBO crystals, this is achieved through the natural birefringence of the material. For example, in type-I SPDC, the pump photon is extraordinary polarized, and the down-converted photons are both ordinary polarized:

$$|\Psi\rangle = \frac{1}{\sqrt{2}} \left(|V\rangle_1 |H\rangle_2 + e^{i\phi} |H\rangle_1 |V\rangle_2 \right) \tag{9}$$

where V and H denote the vertical and horizontal polarization states of the photons, respectively, and ϕ is a phase factor[7].

Focused Pump Light

Focusing the pump light can enhance the efficiency of entangled photon pair generation. The focused pump increases the intensity of the pump light within the crystal, thereby increasing the rate of SPDC. However, it can also lead to an asymmetric broadening of the emission cone and spectral broadening, which might affect the quality of entanglement[8].

Theoretical Model

A theoretical model for the frequency-angular spectrum of entangled photon pairs generated by focused pumping in BBO crystals involves calculating the joint spectral intensity (JSI) and joint spectral amplitude (JSA). The interaction Hamiltonian for type-I SPDC can be written as:

$$H_I = \epsilon_0 \int d^3r \, \chi^{(2)} E_p^{(+)} E_s^{(-)} E_i^{(-)} + H.C. \tag{10}$$

where $\chi^{(2)}$ is the second-order susceptibility tensor describing the crystal's nonlinearity, and $E_p^{(+)}$, $E_s^{(-)}$, and $E_i^{(-)}$ are the positive and negative frequency components of the pump, signal, and idler electric fields, respectively.

4 Data analysis

4.1 Bell's Inequality and Its Derivation

Proposed by John Bell in response to the Einstein-Podolsky-Rosen (EPR) paradox, Bell's work demonstrated that if a local hidden variable theory exists (i.e., a theory where the

behavior of physical systems is predetermined by local hidden variables), certain statistical correlations must obey specific mathematical inequalities. Experimental violations of these inequalities would falsify local hidden variable theories and support the non-locality inherent in quantum mechanics.

Assuming two particles exhibit perfect anti-correlation when measured along the same direction (e.g., at angle θ), Bell's inequality can be expressed as:

$$|P(a,b) - P(a,c)| \leq 1 + P(b,c) \tag{11}$$

Here, $P(a,b)$ represents the correlation between the two particles measured along directions a and b.

The derivation of this formula proceeds as follows: Consider two particles with opposite spin orientations in a three-dimensional space. All possible spin configurations for these particles are summarized in the table below.

Table 2: Spin Configuration Probabilities

Ax	**Ay**	**Az**	**Bx**	**By**	**Bz**	**Probability**
+	+	+	−	−	−	N_1
+	+	−	−	−	+	N_2
+	−	+	−	+	−	N_3
+	−	−	−	+	+	N_4
−	+	+	+	−	−	N_5
−	+	−	+	−	+	N_6
−	−	+	+	+	−	N_7
−	−	−	+	+	+	N_8

Since the total probability must satisfy the normalization condition, we have:

$$\sum_i N_i = 1$$

We assume that the probability of particle A and particle B having the same spin orientation along the x-direction and y-direction, respectively, is denoted by P_{xy}, then

$$P_{xy} = -N_1 - N_2 + N_3 + N_4 + N_5 + N_6 - N_7 - N_8 \tag{12}$$

$$P_{zy} = -N_1 + N_2 + N_3 - N_4 - N_5 + N_6 + N_7 - N_8 \tag{13}$$

$$P_{xz} = -N_1 + N_2 - N_3 + N_4 + N_5 - N_6 + N_7 - N_8 \tag{14}$$

$$|P_{xz} - P_{zy}| = |2(N_4 + N_5) - 2(N_3 + N_6)| \tag{15}$$

$$|P_{xz} - P_{zy}| \leq 2(N_3 + N_4 + N_5 + N_6) \tag{16}$$

$$|P_{xz} - P_{zy}| \leq 1 + P_{xy} \tag{17}$$

4.2 CHSH Inequality and Its Derivation

The original Bell's inequality requires experiments to satisfy the condition of "perfect anti-correlation" (e.g., measurement outcomes of two particles must be strictly opposite when measured along the same direction). In practical experiments, however, this condition is

challenging to satisfy rigorously due to factors such as detector efficiency and experimental noise.

In contrast, the CHSH inequality (Clauser-Horne-Shimony-Holt inequality) does not rely on perfect anti-correlation. Instead, it directly calculates a linear combination of correlation functions based on two sets of measurement direction choices.

The derivation of the CHSH inequality relies on the assumptions of Local Hidden Variable Theory:

4.2.1 Hidden Variable (λ)

It is hypothesized that there exists a hidden variable λ (an unobserved random variable) that carries all unknown information of the system. This variable follows a probability distribution $\rho(\lambda)$, satisfying the normalization condition:

$$\int \rho(\lambda)\, d\lambda = 1. \tag{18}$$

4.2.2 Locality

Alice's measurement outcome depends only on her chosen direction $\mathbf{a}$ and λ, while Bob's outcome depends solely on his direction $\mathbf{b}$ and λ. Formally, this is expressed as:

$$A(\mathbf{a}, \lambda) = \pm 1, \quad B(\mathbf{b}, \lambda) = \pm 1, \tag{19}$$

where $\mathbf{a}$ and $\mathbf{b}$ represent measurement directions (e.g., polarization angles).

4.2.3 Derivation

Suppose two observers, Alice and Bob, participate in the experiment: Alice chooses measurement directions $\mathbf{a}$ and $\mathbf{a}'$, while Bob selects directions $\mathbf{b}$ and $\mathbf{b}'$.

The expectation value $E(a, b)$ for joint measurements along directions a and b can be expressed as:

$$E(a, b) = \int A(a, \lambda) B(b, \lambda)\, \rho(\lambda)\, d\lambda$$

Define four distinct measurement directions $\mathbf{a}, \mathbf{a}', \mathbf{b}, \mathbf{b}'$, and construct the statistical quantity S:

$$S = E(\mathbf{a}, \mathbf{b}) - E(\mathbf{a}, \mathbf{b}') + E(\mathbf{a}', \mathbf{b}) + E(\mathbf{a}', \mathbf{b}')$$

$$S = \int \Big[A(\mathbf{a}, \lambda) B(\mathbf{b}, \lambda) - A(\mathbf{a}, \lambda) B(\mathbf{b}', \lambda) + A(\mathbf{a}', \lambda) B(\mathbf{b}, \lambda) + A(\mathbf{a}', \lambda) B(\mathbf{b}', \lambda) \Big] \rho(\lambda)\, d\lambda \tag{20}$$

$$S = \int \Big[A(\mathbf{a}, \lambda)(B(\mathbf{b}, \lambda) - B(\mathbf{b}', \lambda)) + A(\mathbf{a}', \lambda)(B(\mathbf{b}, \lambda) + B(\mathbf{b}', \lambda)) \Big] \rho(\lambda)\, d\lambda \tag{21}$$

Because $A(\mathbf{a}, \lambda) = \pm 1$, $A(\mathbf{a}', \lambda) = \pm 1$, $B(\mathbf{b}, \lambda) = \pm 1$, and $B(\mathbf{b}', \lambda) = \pm 1$. Consider the term inside the square brackets: $A(\mathbf{a}, \lambda)(B(\mathbf{b}, \lambda) - B(\mathbf{b}', \lambda)) + A(\mathbf{a}', \lambda)(B(\mathbf{b}, \lambda) + B(\mathbf{b}', \lambda))$. If $B(\mathbf{b}, \lambda) = B(\mathbf{b}', \lambda)$, the first term is 0 and the second term is $\pm 1 \times (\pm 2) = \pm 2$. If $B(\mathbf{b}, \lambda) = -B(\mathbf{b}', \lambda)$, the second term is 0 and the first term is $\pm 1 \times (\pm 2) = \pm 2$. In either case, the absolute value of the term inside the brackets is at most 2:

$$|A(\mathbf{a}, \lambda)(B(\mathbf{b}, \lambda) - B(\mathbf{b}', \lambda)) + A(\mathbf{a}', \lambda)(B(\mathbf{b}, \lambda) + B(\mathbf{b}', \lambda))| \leq 2$$

Therefore, integrating over λ:

$$|S| = \left| \int \Big[\dots \Big] \rho(\lambda)\, d\lambda \right| \leq \int \left| \Big[\dots \Big] \right| \rho(\lambda)\, d\lambda \leq \int 2\rho(\lambda)\, d\lambda = 2 \int \rho(\lambda)\, d\lambda = 2$$

So we get the CHSH inequality:

$$|S| = |E(\mathbf{a}, \mathbf{b}) - E(\mathbf{a}, \mathbf{b}') + E(\mathbf{a}', \mathbf{b}) + E(\mathbf{a}', \mathbf{b}')| \leq 2$$

4.3 Data analysis in the experiment

The following is the data from our experiment:

Table 3: Experimental Measurement Results (Coincidence Counts $N(a, b)$)

a (Alice) \b (Bob)	**-22.5°**	**67.5°**	**22.5°**	**112.5°**
$-45°$	6229	1076	1059	6384
$45°$	1086	6203	6222	1069
$0°$	6418	1054	6187	1076
$90°$	1029	6087	1081	6224

4.3.1 Experimental Configuration

Measurement angles chosen for the CHSH test:

- Alice's angles: $a = 0°$, $a' = 45°$
- Bob's angles: $b = 22.5°$, $b' = 67.5°$

4.3.2 Correlation Functions Calculation

Using the formula for the correlation coefficient $E(a, b)$, assuming the counts $N_{\text{same/diff}}$ are correctly identified from the table (NEEDS VERIFICATION based on experimental setup):

$$E(a, b) = \frac{N_{\text{same}}(a, b) - N_{\text{diff}}(a, b)}{N_{\text{same}}(a, b) + N_{\text{diff}}(a, b)}$$

Let's assume the interpretation used in the original code is correct (please verify this mapping with your experiment notes):

$$E(0°, 22.5°) = \frac{6187 - 1076}{6187 + 1076} = \frac{5111}{7263} \approx 0.7037$$
$$E(0°, 67.5°) = \frac{1054 - 6418}{1054 + 6418} = \frac{-5364}{7472} \approx -0.7181$$
$$E(45°, 22.5°) = \frac{6222 - 1069}{6222 + 1069} = \frac{5153}{7291} \approx 0.7070$$
$$E(45°, 67.5°) = \frac{6203 - 1086}{6203 + 1086} = \frac{5117}{7289} \approx 0.7020$$

Self-consistency Check:

Quantum mechanics predicts $E(a, b) = \cos(2(a - b))$ for the singlet state (perfect anti-correlation). $E(0, 22.5) = \cos(2(0 - 22.5)) = \cos(-45°) = 1/\sqrt{2} \approx 0.7071$ (Close

to 0.7037) $E(0, 67.5) = \cos(2(0 - 67.5)) = \cos(-135^\circ) = -1/\sqrt{2} \approx -0.7071$ (Close to -0.7181) $E(45, 22.5) = \cos(2(45 - 22.5)) = \cos(45^\circ) = 1/\sqrt{2} \approx 0.7071$ (Close to 0.7070) $E(45, 67.5) = \cos(2(45 - 67.5)) = \cos(-45^\circ) = 1/\sqrt{2} \approx 0.7071$ (Close to 0.7020) The calculated values are reasonably close to the theoretical QM predictions.

4.3.3 CHSH S-Value Calculation

Using the CHSH combination:

$$S = E(a, b) - E(a, b') + E(a', b) + E(a', b')$$

Substituting values:

$$S = E(0^\circ, 22.5^\circ) - E(0^\circ, 67.5^\circ) + E(45^\circ, 22.5^\circ) + E(45^\circ, 67.5^\circ)$$

$S \approx 0.7037 - (-0.7181) + 0.7070 + 0.7020 = 0.7037 + 0.7181 + 0.7070 + 0.7020 \approx 2.8308$

4.3.4 Experimental Result Comparison

The measured S-value is compared to the classical limit and the quantum prediction for ideal conditions:

- Calculated experimental value: $S_{\text{exp}} \approx 2.8308$
- Quantum mechanical prediction (for the chosen angles): $S_{QM} = \cos(2(a - b)) - \cos(2(a - b')) + \cos(2(a' - b)) + \cos(2(a' - b')) = \cos(-45^\circ) - \cos(-135^\circ) + \cos(45^\circ) + \cos(-45^\circ) = (1/\sqrt{2}) - (-1/\sqrt{2}) + (1/\sqrt{2}) + (1/\sqrt{2}) = 4/\sqrt{2} = 2\sqrt{2} \approx 2.8284$
- Classical limit (Local Realism): $|S| \leq 2$

Final Experimental Result (as stated in original): $S_{\text{exp}} = 2.8332 \pm 0.005$

4.3.5 Conclusion

The experimental result $S_{\text{exp}} = 2.8332 \pm 0.005$ clearly violates the classical CHSH inequality limit of $|S| \leq 2$. The value is consistent with the quantum mechanical prediction of $S = 2\sqrt{2} \approx 2.828$. This provides strong evidence against local hidden variable theories and supports the predictions of quantum mechanics regarding nonlocality.

4.4 Error Analysis

1. **Statistical Uncertainty:** Finiteness of coincidence counts (N) leads to statistical fluctuations. The uncertainty in each count is typically estimated as $\sqrt{N}$, and this uncertainty propagates through the calculation of $E(a, b)$ and S. This is likely the source of the reported ± 0.005 error on S.
2. **Systematic Errors:**
    - **Angle Calibration:** Inaccuracies in setting the polarizer angles (a, a', b, b') can lead to deviations from the theoretical $E(a, b)$ values and affect the calculated S.

- **Detector Efficiency Mismatch:** Differences in efficiency between the detectors for Alice and Bob, or dependence of efficiency on polarization, can affect correlation measurements.
- **Imperfect Entanglement Source:** The source might not produce perfectly entangled states (e.g., state purity less than 1), reducing the observed correlations.
- **Accidental Coincidences:** Random background counts or detector dark counts can lead to false coincidences, affecting the measured N values.
- **Alignment:** Imperfect alignment of the optical components can reduce measurement contrast.

5 Applications

The violation of Bell inequalities enables practical quantum technologies beyond foundational tests. We discuss three key applications, with detailed analysis of quantum teleportation.

5.1 Key Application Areas

5.1.1 Quantum Teleportation

Bell-state measurements using entangled pairs ($S > 2$ violation) allow transmission of unknown quantum states without physical transfer [9]. This constitutes the essential foundation of quantum teleportation. Quantum teleportation is a process that enables the transfer of quantum information from one location to another without physically moving the quantum system itself. It suggests that the correlations between quantum systems are much stronger than any classical correlation could be [10].

5.1.2 Quantum Computing

Bell tests verify cluster state entanglement for measurement-based quantum computation [11]. In measurement - based quantum computation (MBQC), cluster states, highly entangled multi - qubit resources, are used. MBQC conducts calculations via measurements on cluster states, different from the traditional gate - model. Bell tests verify the entanglement properties of cluster states, ensuring accurate measurements in MBQC and its proper operation.

5.1.3 Quantum Metrology

Quantum metrology is a discipline that uses the principles of quantum mechanics to achieve high - precision and high - sensitivity measurements.It takes advantage of unique quantum state properties like superposition and entanglement. Superposition allows a quantum system to be in multiple states simultaneously, such as a qubit representing both 0 and 1. Entanglement creates a strong correlation between quantum particles, where measuring one instantly affects the other regardless of distance. Manipulating these properties helps break through classical measurement limits. For instance, Bell-violating states achieve $\sqrt{N}$ precision enhancement in interferometry [12].

5.2 Quantum Teleportation Protocol

5.2.1 Protocol Steps

1. **Entanglement sharing**: Supppose the particle 1 that Alice wants to teleport is $|\psi\rangle_1 = \alpha|0\rangle_1 + \beta|1\rangle_1$, and the entangled pair of particles 2 and 3 shared by Alice and Bob is in the state
$$|\Psi^-\rangle_{23} = \frac{|01\rangle_{23} - |10\rangle_{23}}{\sqrt{2}} \tag{22}$$
Alice has particle 2 and Bob has particle 3.

2. **Bell measurement**: Alice measures particle 1 ($|\psi\rangle_1 = \alpha|0\rangle_1 + \beta|1\rangle_1$) and her Einstein-Podolsky-Rosen (EPR) pair half (particle 2) in Bell basis. During this process, the initial state of particle 1 is destroyed because of the projection postulate. Once the particle 1 and particle 2 are projected into the Bell basis, the information will be reflected in particle 3. Since the original state of particle 1 is destroyed, it does not violate the no-cloning theorem.

3. **Classical communication**: Alice informs Bob of the Bell basis measurement results through a classical communication channel. Based on the received classical information, Bob performs corresponding unitary transformation operations on the particles he holds, thereby accurately reconstructing the original unknown quantum state.

4. **State recovery**: Based on Alice's measurement result, Bob applies a corresponding unitary operation (e.g. I, X, Z, or XZ gate) to Particle 3 to reconstruct the original state.

5.2.2 Simple Example

Suppose after a specific measurement, particle 1 and particle 2 are projected onto the state:
$$|\Psi^-\rangle_{12} = \frac{|01\rangle_{12} - |10\rangle_{12}}{\sqrt{2}} \tag{23}$$
In this state, when we observe particle 1 and particle 2, particle 2 would always be in the state opposite to particle 1. Since initially, particle 2 and particle 3 are the entangled pair in the state $|\Psi^-\rangle_{23}$, which also exhibits the opposite states after observation, particle 3 will be projected into the initial state of particle 1. Then Bob can get the information Alice wants to teleport by state recovery strategy.

This is only one simple case. In reality, with equal probabilities of 25% , we could find particle 1 and particle 2 in any one of the three other entangled states [10].

5.2.3 Mathematical Description

Bell states are typically represented by the following four states:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \tag{24}$$

$$|\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} \tag{25}$$

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}} \tag{26}$$

$$|\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}} \tag{27}$$

According to the situation mentioned in the "Protocol Steps" section, the complete state of particle 1 and particle 2 before measurement:

$$|\Psi_{total}\rangle = |\psi\rangle_1 \otimes |\Psi^-\rangle_{23} = (\alpha|0\rangle_1 + \beta|1\rangle_1) \otimes \frac{|01\rangle_{23} - |10\rangle_{23}}{\sqrt{2}} \tag{28}$$

After calculation, we can find that:

$$\begin{aligned} |\Psi_{total}\rangle &= \frac{1}{\sqrt{2}}(\alpha|0\rangle_1|01\rangle_{23} - \alpha|0\rangle_1|10\rangle_{23} + \beta|1\rangle_1|01\rangle_{23} - \beta|1\rangle_1|10\rangle_{23}) \\ &= \frac{1}{\sqrt{2}}(\alpha|00\rangle_{12}|1\rangle_3 - \alpha|01\rangle_{12}|0\rangle_3 + \beta|10\rangle_{12}|1\rangle_3 - \beta|11\rangle_{12}|0\rangle_3) \end{aligned} \tag{29}$$

Combining equations above, we can rewrite $|\Psi_{total}\rangle$ in Bell basis $\{|\Phi^\pm\rangle, |\Psi^\pm\rangle\}$:

$$\begin{aligned} |\Psi_{total}\rangle = \frac{1}{2}\Big[&|\Psi^-\rangle_{12}(-\alpha|0\rangle_3 - \beta|1\rangle_3) \\ &+|\Psi^+\rangle_{12}(-\alpha|0\rangle_3 + \beta|1\rangle_3) \\ &+|\Phi^-\rangle_{12}(\alpha|1\rangle_3 + \beta|0\rangle_3) \\ &+|\Phi^+\rangle_{12}(\alpha|1\rangle_3 - \beta|0\rangle_3)\Big] \end{aligned} \tag{30}$$

When Alice measures:

- $|\Psi^-\rangle$ outcome: Bob's state collapses to $-\alpha|0\rangle - \beta|1\rangle$ (requires I correction)
- $|\Psi^+\rangle$ outcome: $-\alpha|0\rangle + \beta|1\rangle$ (requires Z correction)
- $|\Phi^-\rangle$ outcome: $\alpha|1\rangle + \beta|0\rangle$ (requires X correction)
- $|\Phi^+\rangle$ outcome: $\alpha|1\rangle - \beta|0\rangle$ (requires XZ correction)

The CHSH violation ($S = 2\sqrt{2}$) certifies entanglement quality [3].

If Alice measures $|\Psi^-\rangle_{12}$, then the state of particle 3 is $(-\alpha|0\rangle_3 - \beta|1\rangle_3)$, which is the global phase flip of the original state $|\psi\rangle_1$. They are physically equivalent. It is consistent with what we previously discussed in the "Simple Example" section. If Alice measures other states, Bob can eliminate the transformation introduced by the measurement through Pauli gate operations (such as X, Z, XZ) to restore the original state. For example, the matrix corresponding to the gate X is $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. For $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, $X|\psi\rangle = \alpha|1\rangle + \beta|0\rangle$. Thus, when Alice measures $|\Phi^-\rangle_{12}$, Bob can use X gate to get the original state of particle 1.

5.2.4 Experimental Realization

The first experiment on quantum teleportation aimed to transfer the quantum state of a photon [10]. First, entangled photon pairs (2 and 3) were generated via parametric down - conversion in a nonlinear crystal. Then, to project photons 1 and 2 onto a Bell state (specifically $|\psi^-\rangle_{12}$), they were made indistinguishable by superposing them at a beam splitter and detecting coincidences. To ensure they couldn't be distinguished by arrival times, a pulsed pump beam and narrow - bandwidth filters were used.

To verify quantum teleportation, specific measurement bases were chosen, including states linearly polarized at $-45°$ and $+45°$, and circular polarization. The polarization of photon 3 was analyzed to check if it matched the initial polarization of photon 1. Spurious coincidences were identified and subtracted from the data. Overall, the experiment successfully demonstrated the teleportation of a single - photon's quantum state.

5.3 Conclusion

Bell tests transitioned from philosophical tools to enabling technologies: (1) teleportation relies on Bell-state measurement, (2) quantum computing uses Bell violations to verify entanglement, and (3) metrology exploits Bell-inequality-violating states for enhanced precision. The CHSH parameter $S > 2$ quantitatively links foundational physics to practical applications.

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The AI tool served as a helpful assistant, not a replacement for our group's own work and discussion.

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